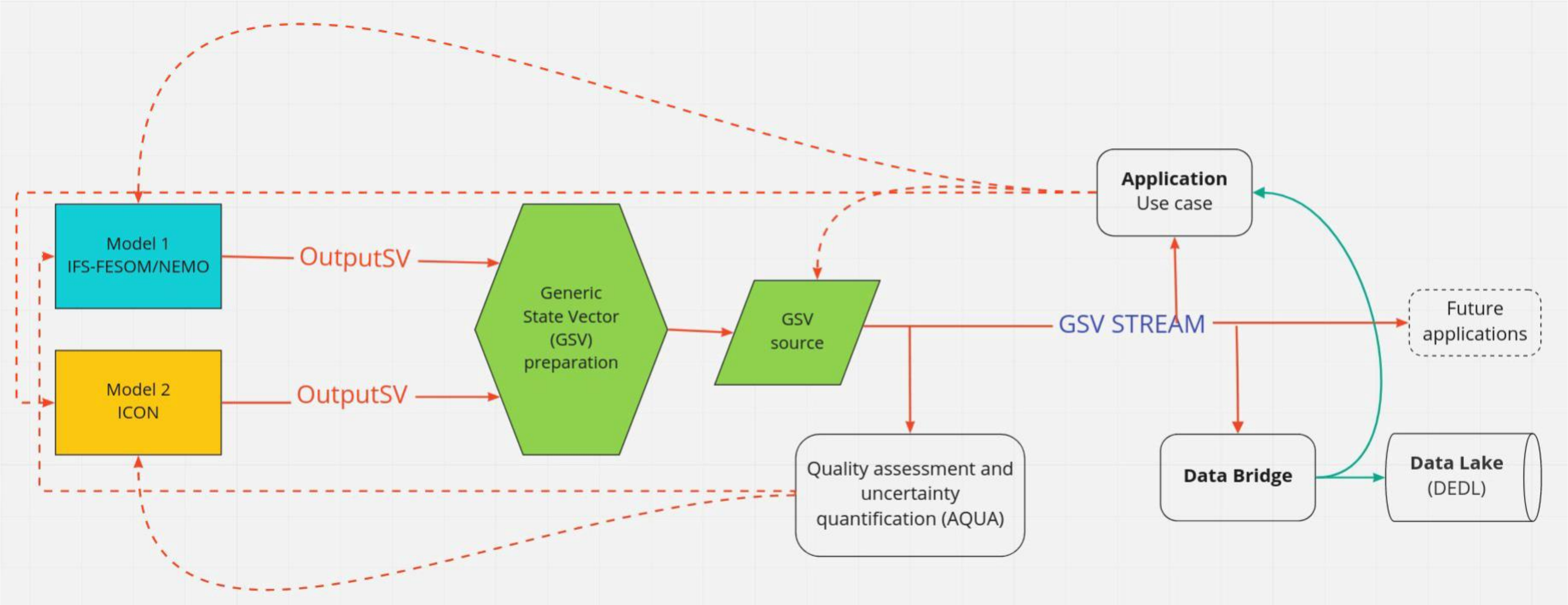
Accessing Climate DT data

Juha Tonttila, Devaraju Narayanappa

Learning outcomes

- Undestand the general technical layout of the data
- Know how to interact with the interfaces to DestinE data
- Understand the basic structure of data requests
- Know how to find the request parameters for a specific experiment and model output variables

ClimateDT workflow – technical architecture



GSV definition

- A **common, standardized data structure** into which outputs from different climate models are transformed, so that they can be uniformly accessed, processed and streamed downstream applications.

More specifically,

- It **represents the full model state** (i.e. Climate variables describing atm, ocn, land etc) at a given time
- It is **generated by climate models** and continuously evolved over time.
- It is **streamed directly to applications** (impact sector models)

- Why ”**Generic**”

  Different climate models (e.g.,ICON, IFS-FESOM,IFS-NEMO)

  - Use **different grids**
  - Have **different metadata conventions**
  - Output **different variable formats**

  GSV solves this by:

  - Coverting all outputs into a **common grid and unified variable structure**
  - Ensuring that **data from different models becomes interoperable**
  - Making datasets **distinguishable only through metadata**, not structure

FDB

- The **Fields Data Base (FDB)** is a system used by ECMWF to store, organize, and retrieve meteorological model data (fields) efficiently.
- Think of FDB as a **fast, structured ”warehouse” for weather/climate models data**, where everything is indexed so you can quickly find exactly what you need.
- Way of organizing GRIB-formatted data

Data Bridge

- Near-compute peripheral infrastructure for data storage, VM and other processing resources
- Climate DT data (currently) resides in Data Bridges deployed to LUMI and Marenostrum5
- *Data Lake* consists of distributed interconnected infrastructures, such as the Data Bridges

Key interfaces

- Data access and processing:
  - Polytope
    - Interface to request Climate DT data from Data Bridges
    - https://polytope.readthedocs.io/en/latest/
  - Earthkit
    - Library consisting of multiple different modules for data access, processing and analysis
    - partially wrapping Polytope functionality
    - https://earthkit.readthedocs.io/en/stable/
    - Particularly for this session: https://earthkit-data.readthedocs.io/en/stable/

Requests

- Similar to ECMWF's MARS request keys
- Data Bridges
  - LUMI: `polytope.lumi.apps.dte.destination-earth.eu`
    - All of ICON, most of IFS-FESOM
  - MN5: `polytope.mn5.apps.dte.destination-earth.eu`
    - All of IFS-NEMO, some of IFS-FESOM

These are fixed:

Key	Relevant values	Description
class	d1	The data originates from Destination Earth
dataset	climate-dt	Selects the Climate DT data from Destination Earth
expver	0001	Experiment version for operational simulations
type	fc	All Climate DT data uses type forecast
generation	2	Climate DT generation

Key	Relevant values	Description
activity	story-nudging, baseline, projections	The type of activity
date	YYYYMMDD (YYYYMMDD/to/YYYYMMDD)	Valid only for clte stream
experiment	cont, hist (baseline or story-nudging), ssp3-7.0 (projections)	Experiment name
levtype	sfc, pl, o2d, o3d, hl, sol	Level type
level	lev1/lev2/lev3	Only for levtype=pl, o3d, hl or sol
model	ifs-fesom, ifs-nemo, icon	Model name
param	paramid (167/235/…)	The variable ID, check documentation
realization	1 or ensemble number	Number of the ensemble member
resolution	high or standard	HEALPix resolution
Stream	clte, clmn	clte: Hourly instantaneous or daily mean values, clmn: monthly mean values
time	HHMM or HHMM/HHMM/HHMM	Select a specific time (Valid only for clte stream)
year	YYYY	Valid only for the clmn stream
month	MM	Valid only for the clmn stream

Ways of working

- All demos, exercises and other material found in Github:
  - https://github.com/DestinE-Climate-DT/user_workshop_finland
  - Instructions and configs for Python environments provided

- Climate DT user guide: https://platform.destine.eu/services/documents-and-api/doc/?service_name=climate-dt-user-guide

Ways of working

- Where can you work with DestinE data?
  - Own laptop/pc: recommended to create a virtual environment and install Polytope, Earthkit and other packages with your method of choice (pip, conda, mamba...)

  - Via DESP -> Insula Code
    - Jupyter platform with key dependencies

  - On HPC
    - Strongly recommended to use stome type of containerization for Python environments!
    - Check what's available for each system
    - Example: lumi-container-wrapper for simple workflows on LUMI
      - https://docs.lumi-supercomputer.eu/software/installing/container-wrapper

Ways of working

- Note about authentication for the Polytope interface
  - Use your DESP credentials
  - Instructions to get a Python script for authentication provided in the workshop Github page
    - Not the only way, but probably the easiest
    - Will create a file with personal token in your home directory
    - Automatically found by the Polytope interface
    - Just run the authentication script and let it prompt the DESP credentials safely
      - Do not write your credentials inside the scripts, do not export to environment variables in shared environments